

Submitted by.....

1. Derive an expression for v_{out} as a function of v_1 and v_2 for the circuit represented in Fig. 1.

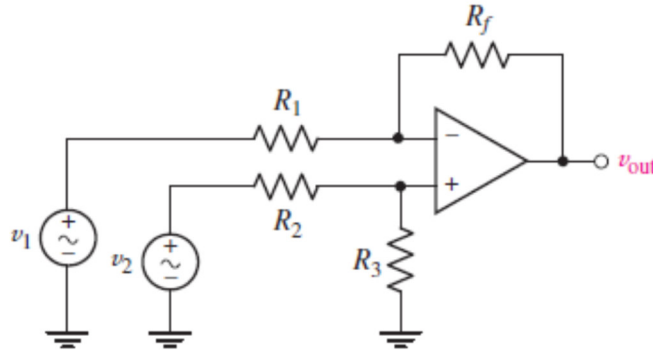


Fig. 1.

2. For the circuit depicted in Fig. 2, calculate v_{out} if $I_s = 2 \text{ mA}$, $R_Y = 4.7 \text{ k}\Omega$, $R_X = 1 \text{ k}\Omega$, and $R_f = 500 \Omega$.

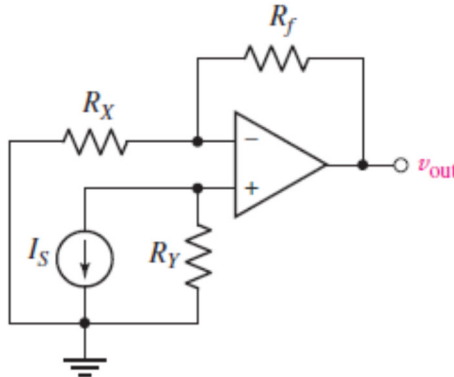


Fig. 2.

3. For the circuit shown in Fig. 3, let $v_{in} = 8 \text{ V}$, and $R_2 = 1 \text{ K}\Omega$, $R_3 = 5 \text{ K}\Omega$, find R_1 to ensure an output voltage $v_{out} = 4 \text{ V}$.

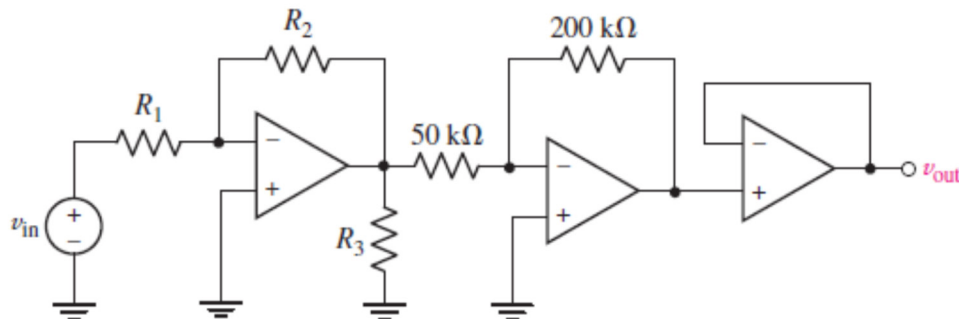


Fig. 3

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4. In the circuit shown in Fig. 4, find k in the voltage transfer function $v_o = kv_s$.

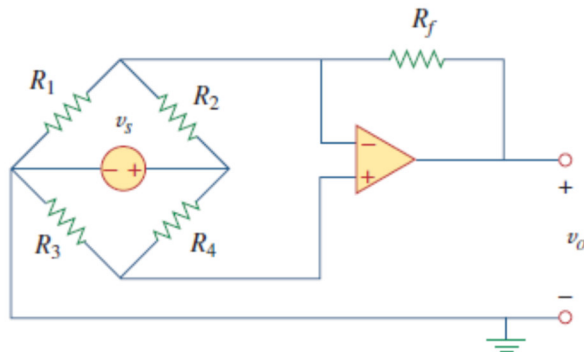


Fig. 4

5. For the circuit in Fig. 5, find i_x .

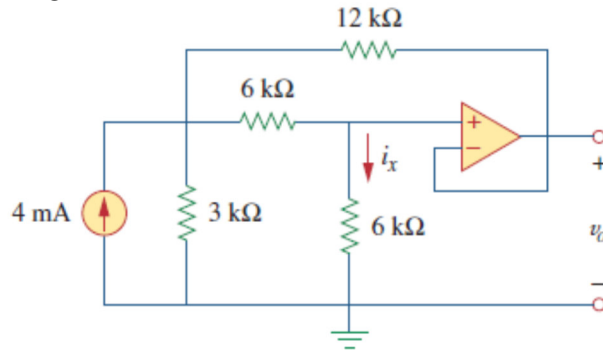


Fig. 5

6. For the circuit of Fig. 6, determine i_x , i_L , and v_L at t equal to
 (a) $t = 0^-$
 (b) $t = 0^+$.

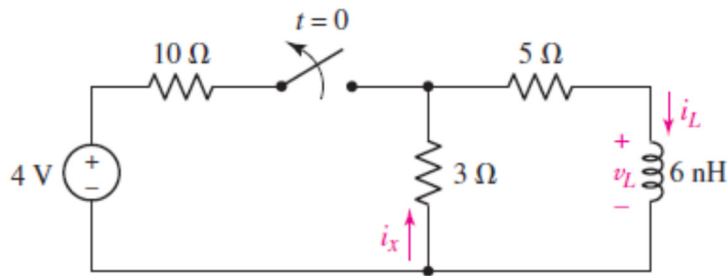


Fig. 6

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7. For the circuit of Fig. 7 , find the voltage labeled v at $t = 200 \mu\text{s}$.

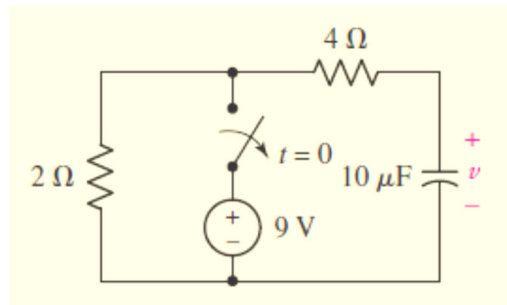


Fig. 7

8. The circuit in Fig. 8 is at steady state before the switch opens. Find the current $i(t)$ for $t > 0$.

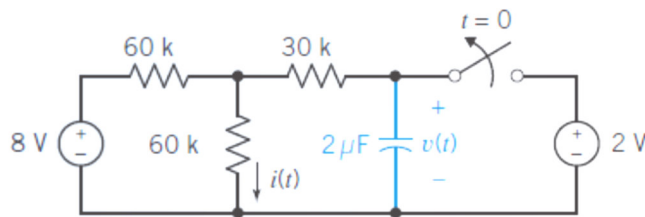


Fig. 8

9. Fig. 9a, shows a plot of the voltage across the inductor in Fig. 9b.

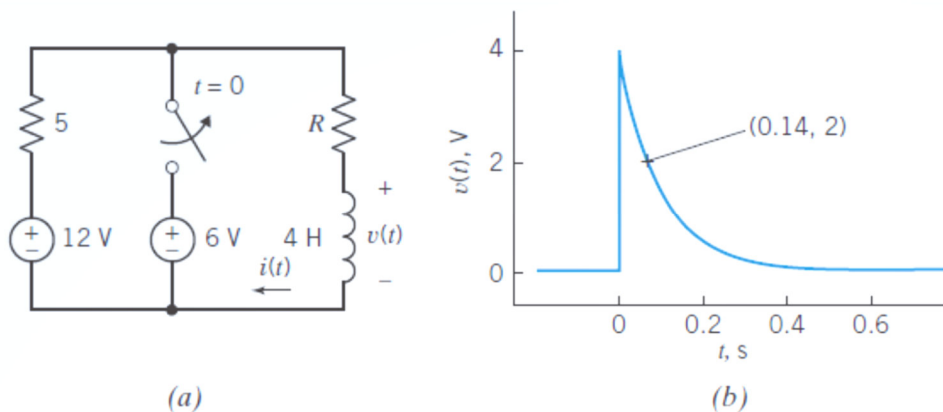


Fig. 9

- Determine the equation that represents the inductor voltage as a function of time.
- Determine the value of the resistance R .
- Determine the equation that represents the inductor current as a function of time.

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10. If $v(0) = 5 \text{ V}$ find $v_o(t)$ for $t > 0$ in the Op Amp circuit in Fig.10. Let $R = 10 \text{ K}\Omega$ and $C = 1 \mu\text{F}$.

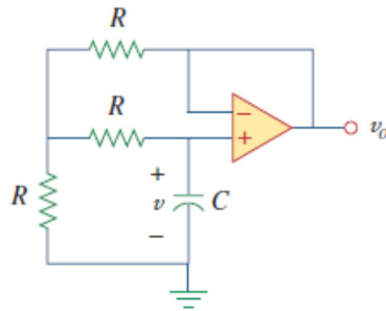


Fig.10